

Solution Architecture

3. Solution Architecture (MVC Design Pattern)

The system enforces strict decoupling of client-server application flows using the Model-View-Controller (MVC) design pattern mapped across the codebase:

Client Browser (React SPA View)



1. REST API Requests (Axios) | 6. Payment Intents (Stripe Elements)

Express Web Server (Node.js Controller & Middleware)



2. Route Guarding (JWT Middleware) | 3. Query & Body Validation

MongoDB Atlas / Mongoose Models / Local uploads (Multer)

1. Frontend (View)

- **Technology Stack:** React.js + Vite + Tailwind CSS.
- **Role:** Serves the dynamic Single Page Application (SPA) to the browser.
- **Key Components:**
 - **Context State Providers:** Maintain user session state ([AuthContext](#)), toast notifications, and persistent shopping cart items.
 - **Product Views:** Render grid search lists with sliders, related product lists, and review feeds.

- **Interactive Wizard Checkout:** Handles shipping location forms, coupon validation flags, and Stripe inputs.

2. Backend (Controller)

- **Technology Stack:** Node.js + Express.js.
- **Role:** Processes incoming RESTful API routes, validates payload bodies, and serves structured JSON responses.
- **Key Controllers:**
 - [authController.js](#): Coordinates registrations, hashes passwords via Bcrypt, and compiles addresses/wishlists.
 - [productController.js](#): Formulates catalog search queries, aggregates stars, and manages review logs.
 - [orderController.js](#): Coordinates stock adjustments, processes payments, and updates fulfillment timelines.
 - [couponController.js](#): Evaluates coupon expiration dates, active toggles, and discounts.

3. Database (Model)

- **Technology Stack:** MongoDB + Mongoose ODM.
 - **Role:** Houses the collections representing the application state.
 - **Models Defined:**
 - [User.js](#): User profiles (name, email, role, wishlist array, saved shipping addresses array).
 - [Product.js](#): Inventory listings (title, description, price, stock quantity, brand, category, rating score, reviews list).
 - [Order.js](#): Financial and tracking items (purchased items, shipping location, shipping rate, tax, total cost, payment status, status tracking timeline).
 - [Coupon.js](#): Discount entries (code, percentage/fixed value, status, expiry dates).
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